

Bhavika Uppala

Data Engineer

3163588451

ubhavika3@gmail.com

Proactive Data Engineer with 5 years of experience designing, building, and optimizing scalable data pipelines and architectures. Proven ability to lead end-to-end data solutions across cloud platforms (AWS, Azure, GCP), big data ecosystems (Spark, Hadoop), and modern data warehouses (Snowflake, Redshift).



Summary Statement

- 5+ years of experience in **architecting, designing, and implementing** end-to-end **ETL pipelines, data integration, and analytics solutions** across **cloud (AWS, Azure, GCP)** and **on-premises** environments.
- Strong hands-on experience with **AWS (S3, Glue, Lambda, Athena, EMR, Redshift, DynamoDB, RDS, Aurora, Kinesis, SQS, SNS, CloudWatch)**, **Azure (Data Lake, Data Factory, Stream Analytics, Site Recovery, Databricks, Azure Marketplace)**, and **GCP (BigQuery, Dataflow, Pub/Sub, Composer, GCS)** for data processing, migration, and analytics.
- Proficient in **Hadoop ecosystem (MapReduce, Hive, HDFS, Distributed Cache, Compression techniques)**, **Spark (PySpark, Structured Streaming, Databricks)**, and **real-time data integration** using **Kafka, Flume, Spark Streaming, and HBase**.
- Extensive experience in **designing and optimizing ETL workflows** using **AWS Glue, Lambda, Athena, Azure Data Factory, GCP Dataflow, and Apache Airflow** with expertise in **Avro, Parquet, JSON, and semi-structured data processing**.
- Strong experience with **Snowflake, Amazon Redshift, Azure Synapse, and Google BigQuery**, including **integrating structured and semi-structured datasets** and **building scalable data marts for business intelligence**.
- Expertise in **CI/CD automation** using **AWS CodePipeline, CodeCommit, CodeBuild, Jenkins, Docker, Kubernetes, GoCD, and Autosys** with a focus on **data pipeline deployment, monitoring, and automation**.
- Worked on **ML lifecycle management** in **Databricks**, **integrating data pipelines with ML models**, and **enabling predictive analytics and real-time insights for business applications**.
- Skilled in **data performance optimization, monitoring (Splunk, AWS CloudWatch)**, **backup, disaster recovery, and troubleshooting in Linux/Unix and Windows environments**.
- Hands-on experience in **relational databases (Oracle, MS SQL Server, MySQL, PostgreSQL)**, **NoSQL databases (DynamoDB, MongoDB, Cassandra)**, and **writing complex SQL/PL-SQL queries for analytics and reporting**.
- Experienced working in **Agile/Scrum and TDD environments**, **collaborating with cross-functional teams**, and **delivering data-driven solutions** to meet evolving business needs.

Work Experience

Azure Data Engineer

Capital One, Richmond, Texas, USA

Jan 2024 - Current

Capital One is a leading American bank specializing in credit cards, auto loans, banking, and savings. I worked on financial data engineering initiatives to modernize cloud-based data platforms, improve data availability, and enable regulatory and risk analytics.

Key Responsibilities:

- Designed, developed, and implemented high-performance **ETL pipelines** using **PySpark (Apache Spark API), Azure Data Factory, and SQL-based transformations**, enabling large-scale data ingestion and processing from diverse structured and semi-structured formats (Parquet, Avro, JSON, XML, CSV, ORC).
- Ingested and processed enterprise-scale data into **Azure Data Lake, Azure Blob Storage, Azure SQL Database, Azure Synapse Analytics (DW), and Databricks**, ensuring secure, scalable, and reliable data flows across on-premises and cloud systems.
- Built **real-time data ingestion pipelines** using **Kafka, Spark Streaming, and Databricks**, applied data quality checks, and delivered clean datasets for **streaming analytics and machine learning workloads**.
- Achieved **70% faster EMR cluster launches** and **60% optimization in Hadoop job processing**; fine-tuned **Elasticsearch clusters** through shard optimization, heap memory management, and query profiling for improved search and analytics performance.

- Partnered with data science teams to prepare **high-quality feature datasets in Databricks**, orchestrated **ML model pipelines**, and supported **real-time predictive analytics** use cases.
- Developed and optimized data models in **Azure Synapse, Snowflake, and Teradata**, leveraging **T-SQL, Python, and Scala** for transformations, cleansing (including SCD Type-2 implementations), and advanced analytics.
- Migrated **on-prem SQL Server workloads to Azure Synapse Analytics and Azure SQL DB**, provisioned high availability in **AWS EC2 & Azure VMs**, and automated infrastructure with **Terraform, Chef, and Kubernetes**, improving scalability by 20%.
- Implemented **CI/CD pipelines** using **Jenkins, GitHub, Maven, and Terraform** for automated deployment of data pipelines and infrastructure, reducing deployment errors and accelerating release cycles.
- Worked across all phases of **SDLC**—requirements gathering, design, development, deployment, and production support—while collaborating with cross-functional teams in Agile environments.
- Built **PySpark validation frameworks** for raw-to-curated data quality checks in **Snowflake and Azure Synapse**, developed **SSIS/DTS packages** for ETL, and designed dashboards/reports for data-driven decision-making.

Technologies Used: Azure, Azure Data Lake, Azure Synapse Analytics, CI/CD, Data Factory, EC2, Elasticsearch, EMR, ETL, Factory, Git, Hive, Java, Jenkins, JS, Kafka, Kubernetes, PySpark, Python, S3, Scala, Services, Snowflake, Spark, Spark

AWS Data Engineer HealthFirst Medical Group, Dallas, Texas, USA Mar 2023 – Dec 2023

HealthFirst Medical Group is a mid-sized healthcare provider offering clinical services and insurance plans. I supported their digital transformation journey by migrating on-prem systems to cloud and enabling analytics-driven decision-making.

Key Responsibilities:

- Designed and developed **ETL pipelines** using **Azure Data Factory (ADF)** and **AWS Glue** to integrate EHR, claims, and patient records from multiple sources.
- Built a centralized **Data Lake** on **Azure ADLS Gen2** and **Amazon S3**, ensuring partitioning and metadata management for scalable access.
- Developed **real-time ingestion pipelines** using **Kafka** and **Kinesis Streams** for patient appointment and pharmacy transactions.
- Modeled data in **Snowflake** and **Azure Synapse** to support population health management and regulatory reporting.
- Created automated **data quality frameworks** using **PySpark** for raw-to-curated validation (null checks, duplicates, schema mismatch).
- Implemented **SCD Type-2** and **CDC (Change Data Capture)** pipelines for maintaining historical patient records.
- Automated infrastructure using **Terraform, CloudFormation**, and **Azure DevOps CI/CD pipelines**, reducing manual efforts by 40%.
- Partnered with analytics teams to provide curated datasets for **predictive risk analysis** and clinical outcome forecasting.
- Ensured compliance with **HIPAA** and internal governance through **data encryption, RBAC, and audit logging**.
- Provided production support, including incident resolution, performance tuning, and pipeline optimization.

Technologies Used: AWS, BigQuery, CI/CD, Cloudera, Docker, Elasticsearch, ETL, Git, Glue, HDFS, Jenkins, Jira, Kafka, Kubernetes, lake, Lambda, Linux, Oozie, Oracle, Pig, S3, SAS, Selenium, Spark, Spark SQL, Spark Streaming, SQL, Sqoop

Data Engineer ICICI Lombard, Chennai, India Aug 2021 – June 2022

ICICI Lombard is a leading general insurance company in India. I worked on building insurance data pipelines, creating reporting systems, and supporting regulatory compliance.

Key Responsibilities:

- Developed **ETL workflows** using **Informatica PowerCenter** and **SQL Server** for policyholder and claims datasets.
- Built **data warehouse schemas** (star and snowflake) for underwriting and premium reporting.
- Ingested external broker and partner datasets in semi-structured formats (Excel, CSV, JSON).
- Designed **real-time dashboards** in **Tableau** for claims tracking and customer support teams.
- Implemented **data validation checks** for accuracy, duplication, and integrity.
- Migrated legacy flat-file systems into SQL-based models for centralized analytics.
- Automated job scheduling using **Autosys** for ETL orchestration.
- Partnered with finance teams to deliver **loss ratio and premium forecasting reports**.
- Implemented **audit logging and lineage tracking** for IRDA compliance.
- Supported production environments by troubleshooting ETL job failures and optimizing performance.

Technologies Used: Informatica PowerCenter, SQL Server, Tableau, Autosys, ETL, Data Warehousing, Audit Logging, IRDA compliance

Data Engineer Max Bupa Health Insurance, Bangalore, India Jan 2021 – June 2021

Max Bupa is a leading Indian health insurance provider. I engineered pipelines and analytics solutions to streamline claims processing and enable risk management.

Key Responsibilities:

- Designed and implemented **enterprise-scale ETL pipelines** using **Informatica** and **SQL Server** to consolidate policyholder, premium, and claims data across multiple lines of business.
- Migrated fragmented on-prem **Oracle and flat-file systems** into a centralized data warehouse, improving query performance by 50% and reducing reporting delays.
- Built **claims fraud detection pipelines** using **Kafka + SQL-based business rules**, enabling real-time anomaly detection and reducing fraudulent payouts by ~20%.
- Partnered with actuarial teams to provision curated datasets for **risk modeling, underwriting, and premium forecasting**, enabling data-driven pricing strategies.
- Developed **customer segmentation and retention data marts**, supporting targeted cross-sell/upsell campaigns and improving customer lifetime value.
- Implemented **SCD Type-2 and CDC frameworks** to maintain historical snapshots of policyholder data for regulatory and audit purposes.
- Designed **interactive dashboards** in **Tableau** for claims, policy lapse rates, and customer churn insights, used by senior management for strategic planning.
- Automated ETL scheduling and monitoring with **Control-M**, ensuring reliable nightly data refreshes across 30+ critical jobs.
- Applied **data governance, audit logging, and encryption standards** to meet **IRDA compliance** for insurance data confidentiality.
- Partnered with cross-functional teams (finance, actuarial, and claims) in **Agile sprints**, ensuring timely delivery of analytics use cases and business reporting.

Technologies Used: Informatica, SQL Server, Oracle, Kafka, Tableau, Control-M, ETL, Data Marts, SCD Type-2, CDC, IRDA compliance

Relevant Skills

- **Cloud Platforms:** AWS (S3, Glue, Lambda, EMR, Redshift, Kinesis, CloudFormation, CloudWatch), Azure (Databricks, Synapse, Data Factory, Data Lake, Stream Analytics, Azure DevOps), GCP (BigQuery, Dataflow, Pub/Sub, GCS)
- **Big Data & Streaming:** Apache Spark (PySpark, Scala, Spark-SQL, Structured Streaming, MLlib), Hadoop (HDFS, Hive, MapReduce), Kafka, Kinesis, Flume, Delta Lake
- **Data Warehousing & BI:** Snowflake, Azure Synapse, Redshift, Teradata, BigQuery, Informatica PowerCenter, Tableau (dashboards, reporting), Data Marts, SCD Type-2, CDC
- **Machine Learning & Analytics:** Feature Engineering, Data Preprocessing, Model Pipeline Orchestration in Databricks, Predictive Analytics, Risk/Fraud Detection Use Cases
- **Databases:** SQL (PostgreSQL, SQL Server, Oracle, MySQL), NoSQL (MongoDB, Cassandra, DynamoDB, HBase), advanced SQL/PL-SQL queries for analytics and reporting
- **Data Orchestration & Scheduling:** Apache Airflow, Oozie, Autosys, Control-M, Terraform, Chef
- **DevOps & CI/CD:** Jenkins, GitHub, Maven, Terraform, Kubernetes, Docker, Azure DevOps Pipelines, AWS CodePipeline/CodeCommit/CodeBuild
- **Testing & Automation:** PyTest, PySpark Validation Frameworks, SSIS/DTS, Selenium
- **Programming Languages:** Python, SQL, Scala, Java

Education Details

- **Masters in Computer Science** from Wichita State University